# MEDLYTIC

## Project Synopsis

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# MEDLYTIC

"Find your medicine, instantly."

## 1. Introduction

Medlytic is an AI-powered web application developed to help users quickly find the availability of medicines at nearby pharmacies. The system allows users to search for medicines, compare prices, locate nearby pharmacies, and receive AI-based suggestions for alternative medicines when the required medicine is unavailable. The application also provides pharmacy ratings, operating hours, and medicine information to improve accessibility and save users time during emergencies. The project aims to reduce the difficulty of searching for medicines manually and improve healthcare convenience through intelligent technology.

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## 2. Objectives of the Project

- To develop a centralized platform for checking medicine availability in nearby pharmacies.
- To help users compare medicine prices across different pharmacies.
- To provide AI-based alternative medicine suggestions when medicines are unavailable.
- To display nearby pharmacies with ratings, contact details, and operating hours.
- To reduce the time spent searching for medicines during emergencies.
- To improve accessibility and convenience through a user-friendly web application.

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## 3. Problem Statement

Many people waste valuable time visiting multiple pharmacies to find the medicines they need, especially during emergencies. There is no centralized platform that allows users to check realtime medicine availability, compare prices, or receive intelligent recommendations for alternative medicines. This often leads to inconvenience, delayed treatment, and increased healthcare costs. The project aims to solve these problems by developing an AI-powered medicine availability finder that connects users with nearby pharmacies and provides smart healthcare assistance**.**

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## 4. Scope of the Project

The system can be used by patients, caregivers, and pharmacies. Users can search for medicines, check availability, compare prices, locate nearby pharmacies, and receive AI-based medicine recommendations. Pharmacies can update medicine stock information to maintain accurate availability. Future enhancements may include online medicine booking, home delivery integration, digital prescriptions, appointment scheduling, and multilingual support.

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## 5. Methodology

The project will follow the Software Development Life Cycle (SDLC):

1. Requirement Analysis
2. System Design
3. Development
4. Testing
5. Implementation
6. Maintenance

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## 6. Hardware and Software Requirements

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 8 GB minimum
- Hard Disk: 256 GB SSD or above

Software Requirements

- Frontend: React.js
- Backend: Node.js with Express.js
- Database: MongoDB
- AI Integration: OpenAI API or Google Gemini API
- Maps API: Google Maps API
- Authentication: Firebase Authentication
- Development Tool: Visual Studio Code

- Hosting: Vercel / Render

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## 7. Expected Outcome

The system will enable users to quickly locate medicines available at nearby pharmacies, compare prices, and receive AI-powered recommendations for alternative medicines. It will reduce the time required to search for medicines, improve access to healthcare services, and provide a reliable platform for patients and pharmacies. The application is expected to enhance user convenience, support informed decision-making, and contribute to more efficient healthcare management.

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## 8. Conclusion

Medlytic is an AI-powered healthcare solution designed to simplify the process of finding medicines. By integrating pharmacy information, real-time availability, price comparison, and intelligent recommendations into a single platform, the system improves accessibility and reduces delays in obtaining essential medicines. The project demonstrates the practical application of Artificial Intelligence in healthcare while offering a user-friendly and efficient solution for everyday medical needs.

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## 9. References


1. **OpenAI API Documentation – https://platform.openai.com/docs**
2. **Google Gemini API Documentation – https://ai.google.dev/**
3. **Google Maps Platform Documentation – https://developers.google.com/maps**
4. **Firebase Documentation – https://firebase.google.com/docs**
5. **React Documentation – https://react.dev/**
6. **Node.js Documentation – https://nodejs.org/docs/**
7. **MongoDB Documentation – https://www.mongodb.com/docs/**